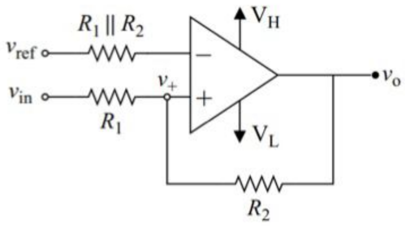
Question 1

For the Given Schmitt Trigger Circuit, V_H = 10V, V_L = -10V, V_{ref} = 1V



(a)	Is this an Inverting or non-inverting Schmitt Trigger?	1
(b)	Let R_1 = 1 kΩ, R_2 = 2 kΩ; Draw V_{out} vs. V_{in} graph (VTC) or Transfer curve with appropriate labels.	2
(c)	Let input voltage be 18V peak to peak sine wave. Draw V_{out} vs. time plot for the given input.	2

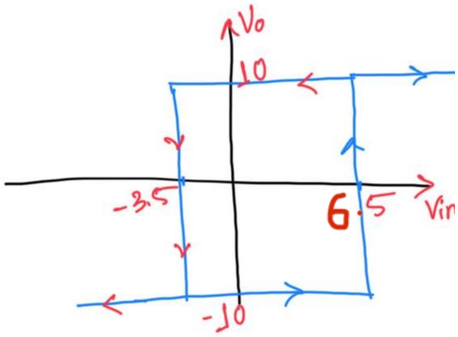
(a) Non-Inverting

(b) $V_{UT} = \left(-\frac{R_1}{R_2}\right)V_L + \frac{R_1+R_2}{R_2}V_{ref}$

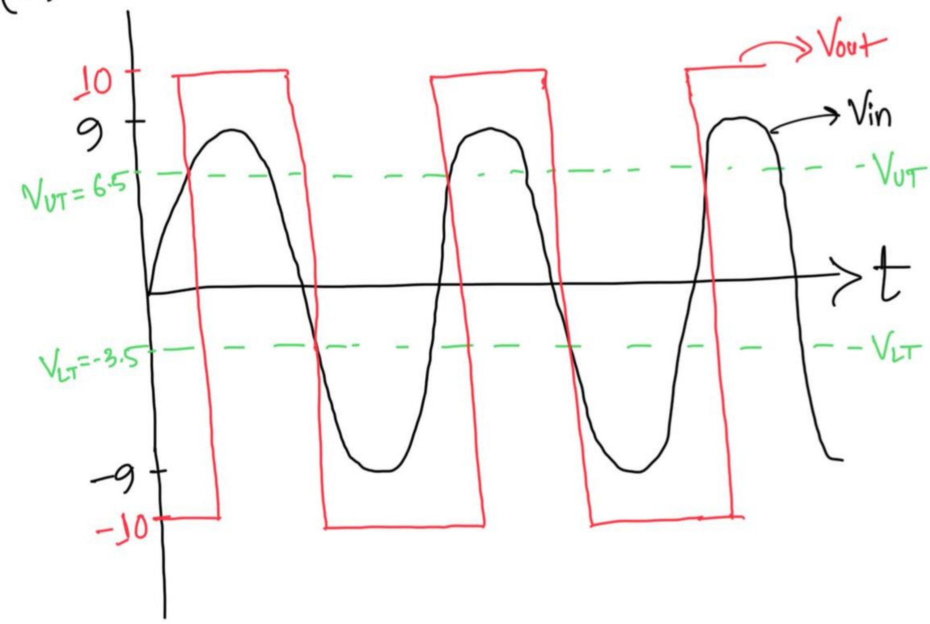
$= -\frac{1}{2}(-10) + \frac{3}{2} \times 1$

$= 5 + 1.5 = 6.5$

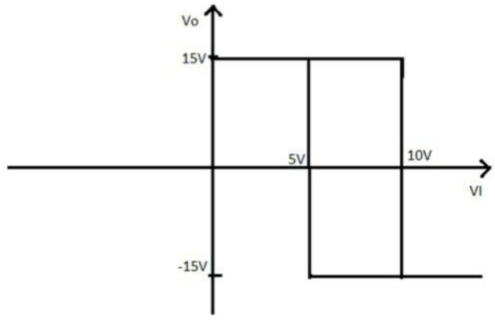
$V_{LT} = -5 + 1.5 = -3.5$



(c)



Question 3



(a)	Identify the Schmitt trigger circuit from the VTC and also find out V_H, V_L, V_{UT}, V_{LT}, V_{shift}	2.5
(b)	Design a Schmitt trigger circuit that will produce the same VTC.	2.5

7 a) This is inverting schmitt trigger with V_{REF}.

7.b $V_H = 15V$,

$V_L = -15V$

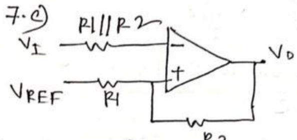
$V_{TH} = +10V$

$V_{TL} = +5V$

$V_{Hysteresis} = V_{TH} - V_{TL} = 5V$

$V_S = \frac{V_{TH} + V_{TL}}{2} = \frac{10+5}{2} = 7.5$

7.c)



$V_S = V_{REF} \times \frac{R2}{R1+R2} \rightarrow (i)$

$V_{Hysteresis} = (V_H - V_L) \frac{R1}{R1+R2} \rightarrow (ii)$

We know, from graph

$V_S = 7.5V$

$V_{Hysteresis} = 5V$

From, (ii), $5 = (15+15) \frac{R1}{R1+R2}$

$\frac{R1}{R1+R2} = \frac{1}{6} \Rightarrow \frac{R1}{R2} = \frac{1}{5}$

Now suppose, $R1 = 1k\Omega$,

then, $\frac{1k}{R_2} = \frac{1}{5}$

$R2 = 5k\Omega$

From (i), $7.5 = V_{REF} \times \frac{5}{1+5}$

$\Rightarrow V_{REF} = +9V$

[redacted] Qus - 3

(a) $R_1 = R_2$, $V_H = 8V$, $V_L = -8V$, $C_x = 100nF$

$f = 200Hz$, $T = \frac{1}{200} = 5ms$, $D = 0.5$

$\therefore T_1 = DT = 2.5\ ms$

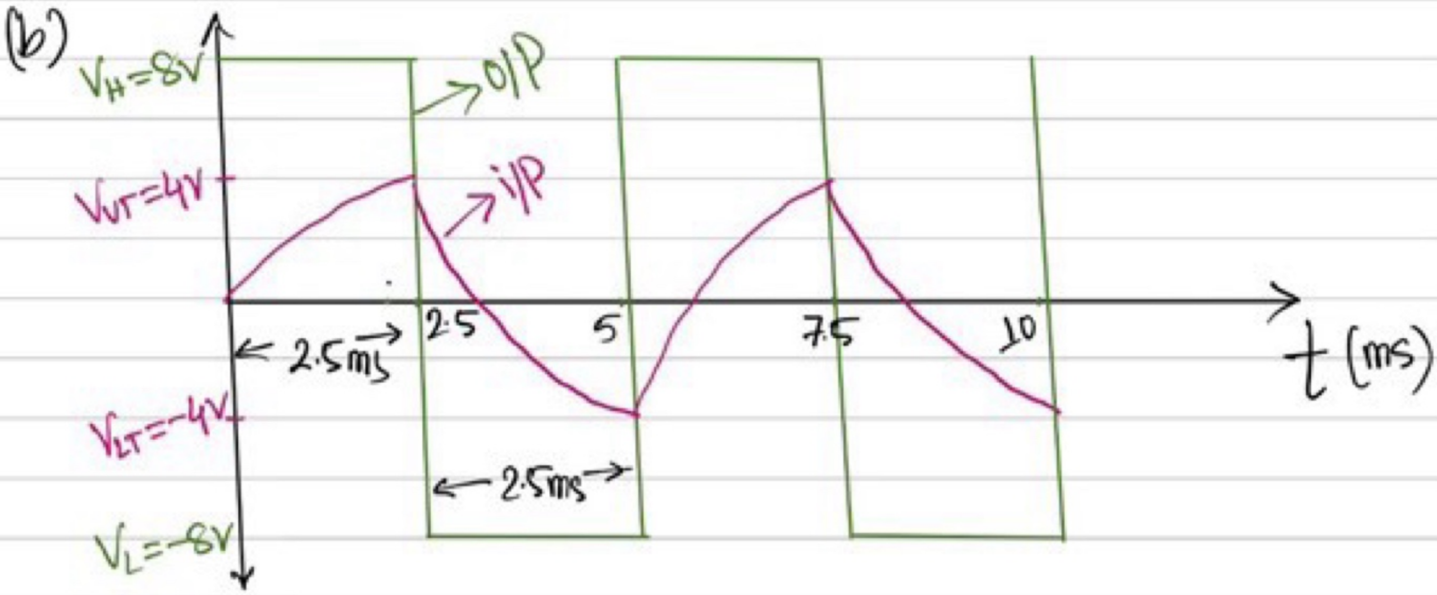
$V_{UT} = \frac{R_1}{R_1+R_2} V_H = 4V$, $V_{LT} = \frac{R_1}{R_1+R_2} V_L = -4V$

$T_1 = R_x C_x \ ln\left|\frac{V_H - V_{LT}}{V_H - V_{UT}}\right|$

$2.5\times10^{-3} = R_x \times 100\times10^{-9}\ ln\left|\frac{8+4}{8-4}\right|$

$\boxed{R_x = 22.75\ KHz}$, Time const, $\tau = R_x C_x = 2.27\ ms$

(b)



(c) here $V_L = -V_H$

$V_{UT} = -\frac{R_1}{R_2} V_L = \frac{-V_L}{8} = \frac{V_H}{8}$

$V_{LT} = -\frac{R_1}{R_2} V_H = \frac{-V_H}{8}$

$T_1 = RC\left(\frac{V_{UT} - V_{LT}}{V_H}\right) = RC\left(\frac{V_H/8 + V_H/8}{V_H}\right) = \frac{RC}{4}$

$T_2 = RC\left(\frac{V_{LT} - V_{UT}}{V_L}\right) = RC\left(\frac{-V_H/8 - V_H/8}{-V_H}\right) = \frac{RC}{4}$

$\therefore T = T_1 + T_2 = \frac{RC}{2}$ $\therefore$ $\boxed{\text{Freq, } f = \frac{1}{T} = \frac{2}{RC}}$

Qstn-4 soln ([redacted])

(a) Triangular Wave

here $V_L = -V_H$, $p = \frac{40}{10} = 4$

$$f = \frac{p}{4R_iC} = \frac{4}{4 \times 2K \times 5\mu f} = 100\ Hz$$

$$T = \frac{1}{f} = 10\ ms$$

(b) From Fig-2 → $V_H = 8V$, $V_L = -16V$, $V_{TH} = 5V$, $V_{TL} = -10V$, $T_1 = 0.25ms$

$$V_{TH} = \frac{R_1}{R_1+R_2} V_H \quad \text{or,} \quad V_{TL} = \frac{R_1}{R_1+R_2} V_L$$

$$5 = \frac{R_1}{R_1+R_2} \times 8 \qquad -10 = \frac{R_1}{R_1+R_2}(-16)$$

$$\Rightarrow \frac{5}{8} = \frac{R_1}{R_1+R_2} \qquad \frac{10}{16} = \frac{R_1}{R_1+R_2}$$

$$\Rightarrow \frac{5}{3} = \frac{R_1}{R_2} \qquad R_1 = \frac{5}{3} R_2$$

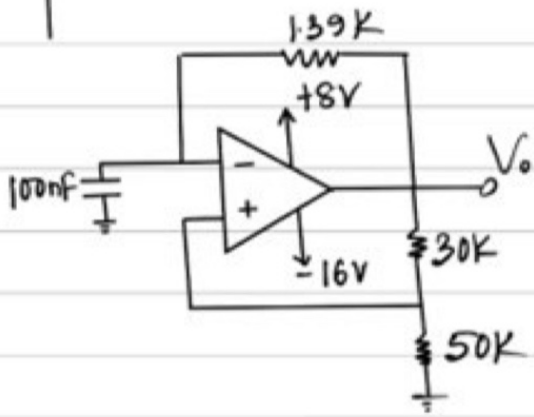
$$\therefore R_1 = \frac{5}{3} R_2$$

$$\boxed{\begin{array}{l} \text{let } R_2 = 30K \\ \therefore R_1 = 50K \end{array}}$$

$$T_1 = R_x C_x \ln\left|\frac{V_H - V_{TL}}{V_H - V_{TH}}\right|$$

$$0.25ms = R_x \times 100nf \times 1.79$$

$$\boxed{R_x = 1.39K}$$



(c) $T_1 = 0.25ms$

$$T_2 = R_x C_x \ln\left|\frac{V_L - V_{TH}}{V_L - V_{TL}}\right|$$

$$= 1.396K \times 100nf \times \ln\left|\frac{-16-5}{-16+10}\right|$$

$$T_2 = 0.17\ ms$$

$\therefore$ Period, $T = T_1 + T_2 = 0.42\ ms$

freq. $f = \frac{1}{T} \approx 2381\ Hz$

Duty cycle, $D = \frac{T_1}{T} \times 100\%$

$$= \frac{0.25}{0.42} \times 100\%$$

$$= 59.52\%$$